Task T4: Update Defender Definitions and Run a Quick Scan

Task: T4

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1. Aim

To write and execute a PowerShell script that updates Windows Defender definitions, performs a quick antivirus scan, and displays the last scan completion date and a result summary[cite: 1, 2].

2. Course Outcome and Module

Course Outcome**:** CO1 – Configure and manage system protection tools and evaluate their ability to defend against threats.

Module: System Protection and Threat Prevention.

3. Tools and Versions

Tool	Exact Version	Operating System
PowerShell	5.1 (Built-in)	Windows 10 / Windows 11
Defender Cmdlets	Built-in Module	Windows 10 / Windows 11

4. Program or Commands

```
# Update Windows Defender Signatures
Update-MpSignature

# Run a quick antivirus scan
Start-MpScan -ScanType QuickScan

# Get Defender status and display summary
$status = Get-MpComputerStatus
Write-Host "-------------------------------------------------------------------------------"
Write-Host "Last Quick Scan Time       : " $status.QuickScanAge "day(s) ago /
    " $status.QuickScanEndTime
Write-Host "Signature Update Time      : " $status.
   AntivirusSignatureLastUpdated
Write-Host "Threats Detected (Count)   : "  $status.ThreatCount
Write-Host "-------------------------------------------------------------------------------"
```

5. How to Run

1. Open PowerShell as an Administrator
2. Set execution policy for the current session

```
Set-ExecutionPolicy -Scope Process Bypass
```

3. Run the script file

```
.\defender_scan.ps1
```

6. Sample Output with Screenshots

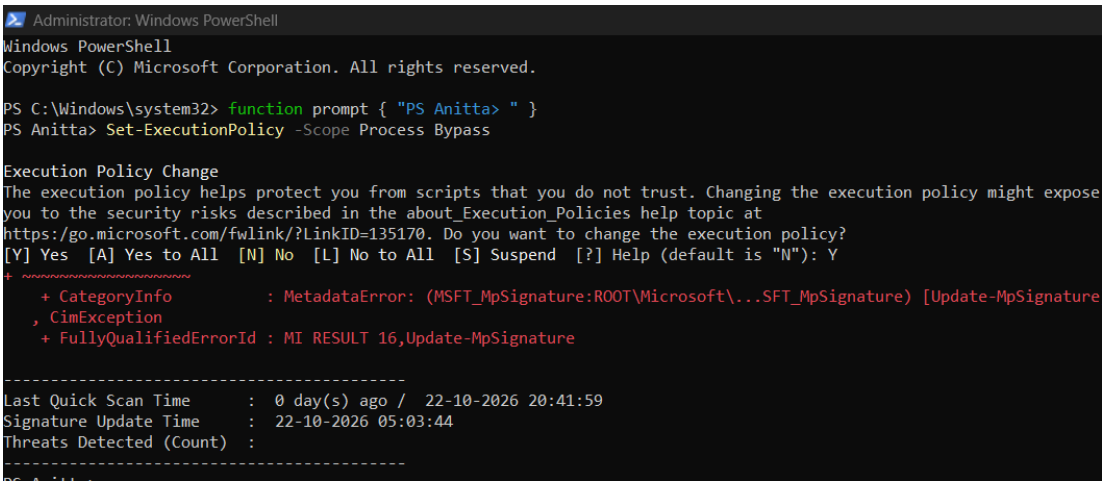


Figure 1: Figure T4-1: Execution output showing Defender signature update, quick scan status, and prompt identity.

7. Result

The script successfully updated the Windows Defender signatures, executed a quick scan, and printed the last scan completion timestamp along with the detected threat count

8. How This Version Differs from Other Versions

Version	Key Differences
PowerShell Script (T4)	Automates signature updates and antivirus scanning via command-line cmdlets (Update-MpSignature, Start-MpScan) without opening a GUI
Windows Security GUI	Requires manual click navigation through system menus to trigger updates and quick scans.

9. Viva: Five Questions with Answers

1. Q: What does the cmdlet Update-MpSignature do?
   It updates the virus and spyware definition signatures used by Microsoft Defender
2. Q: How do you trigger a quick scan versus a full scan in PowerShell?
   Use Start-MpScan -ScanType QuickScan for quick scan and -ScanType FullScan for a full system scan
3. Q: Which cmdlet retrieves the current status of Windows Defender?
   The Get-MpComputerStatus cmdlet
4. Q: Why is -Scope Process Bypass used when changing execution policy?
   It allows script execution temporarily only for the current PowerShell window without altering system-wide policy.
5. Q: Which property returned by Get-MpComputerStatus shows the number of detected threats?
   The ThreatCount property

10. Common Errors: Message, Cause, Fix

- Message: Access is denied / Service cannot be configured
  - Cause: Executing commands without administrative privileges.
  - Fix: Open PowerShell using ”Run as administrator”
- Message: HRESULT 0x800106ba / NotSpecified
  - Cause: Windows Defender engine was disabled by a third-party antivirus (e.g., Reason Cybersecurity / RAV).
  - Fix: Uninstall the third-party antivirus and restart the PC to re-enable Microsoft Defender.

Declaration: AI tools were utilized for LaTeX document structuring, grammatical review, comparison of implementation versions, and assistance in preparing and verifying viva voice questions.